

Requirement Gathering

Project Name

Smart Lender – Loan Eligibility Prediction System

Introduction

Requirement gathering is an important phase of software development that helps identify the needs and expectations of stakeholders before designing and implementing the system.

The Smart Lender project focuses on developing a machine learning-based system that predicts loan eligibility by analyzing applicant information.

Stakeholders Identification

1. Banking Institutions

Banks require an efficient system to analyze loan applications and support faster decision-making.

2. Credit Officers

Credit officers need a system that reduces manual verification effort and provides reliable predictions.

3. Loan Applicants

Applicants expect faster and transparent loan approval decisions.

4. System Administrators

Administrators manage the application, model updates, and system maintenance.

Existing System Challenges

Traditional loan approval systems face several challenges:

- Manual verification takes more time.
 - Large application volumes increase workload.
 - Human errors may affect decisions.
 - Risk assessment depends heavily on manual analysis.
 - Processing delays affect customer experience.
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Proposed System Requirements

The Smart Lender system should:

- Accept applicant details through a web interface.
 - Process and preprocess input data.
 - Apply trained machine learning models.
 - Predict loan approval status.
 - Display prediction results instantly.
 - Provide accurate and consistent decision support.
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Data Requirements

The system requires applicant information including:

- Gender
 - Marital Status
 - Dependents
 - Education
 - Self Employment Status
 - Applicant Income
 - Co-applicant Income
 - Loan Amount
 - Loan Term
 - Credit History
 - Property Area
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Machine Learning Requirements

The system uses:

- Data preprocessing techniques
- Feature encoding
- Missing value handling
- SMOTE data balancing
- Feature scaling
- Classification algorithms

Models considered:

- Decision Tree

- Random Forest
- KNN
- Gradient Boosting

Conclusion

Requirement gathering helped identify the objectives, users, data requirements, and technical needs of the Smart Lender system. These requirements provide a foundation for developing an effective machine learning-based loan prediction application.